

SMART INDIA HACKATHON 2024



★ Problem Statement ID - 1656

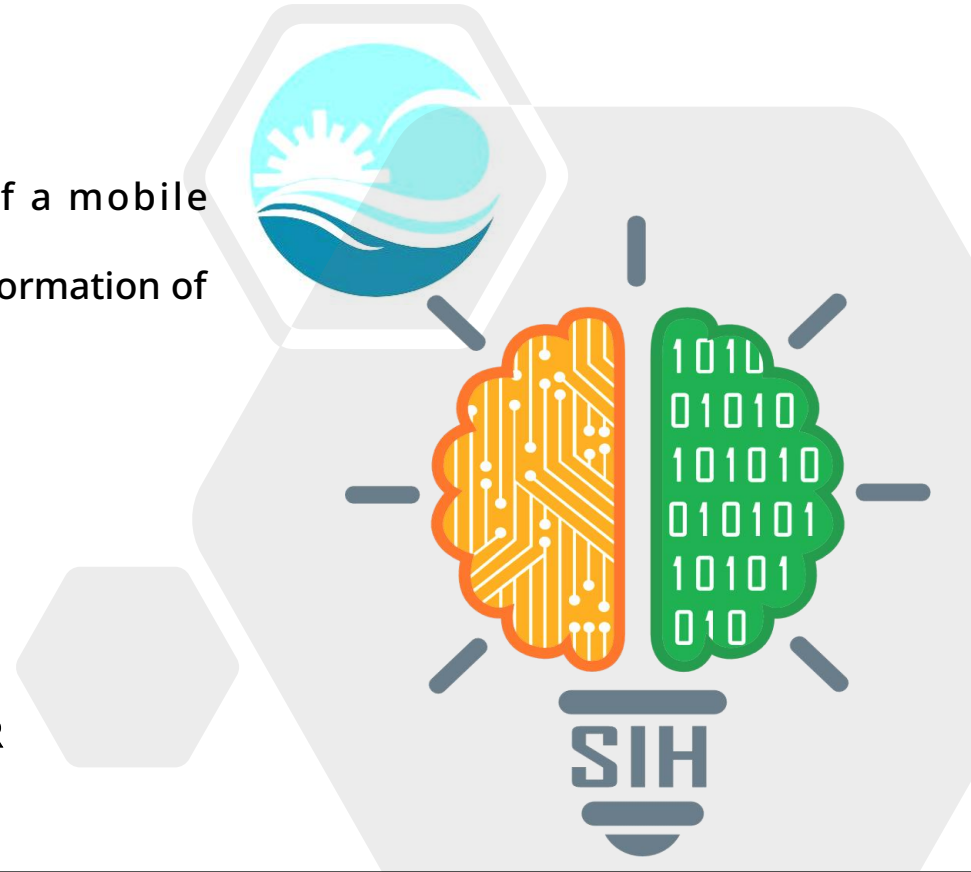
★ Problem Statement Title - Development of a mobile application to provide recreational suitability information of beach locations across India.

★ Theme - Travel & Tourism

★ PS Category- Software/Hardware - Software

★ Team ID - 32490

★ Team Name (Registered on portal) - DEEP WATER





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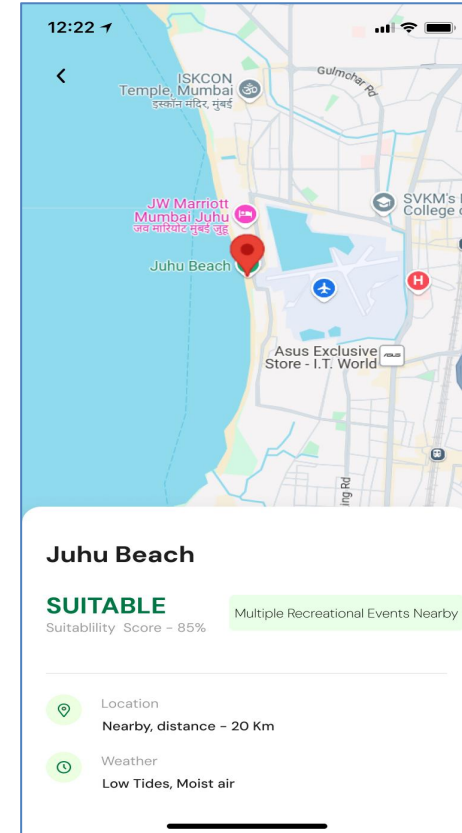
BEACH SAFETY & RECREATIONAL SUITABILITY ANDROID APP VARUNAM



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★ Proposed Solution

- Proposing Solution: A mobile application that provides **real-time suitability information** for recreational activities on beaches across India.
- Addressing the Problem: Enhances tourist safety by providing real-time information about beach conditions, **Alerts users about various hazardous conditions** and supports the government's initiative to promote Tourism.
- Innovation and uniqueness: A first-of-its-kind solution focused on **coastal recreational suitability** with a national coverage of beaches, **Multilingual support** and **interactive geo-spatial map interface** with **push-alert notifications**.

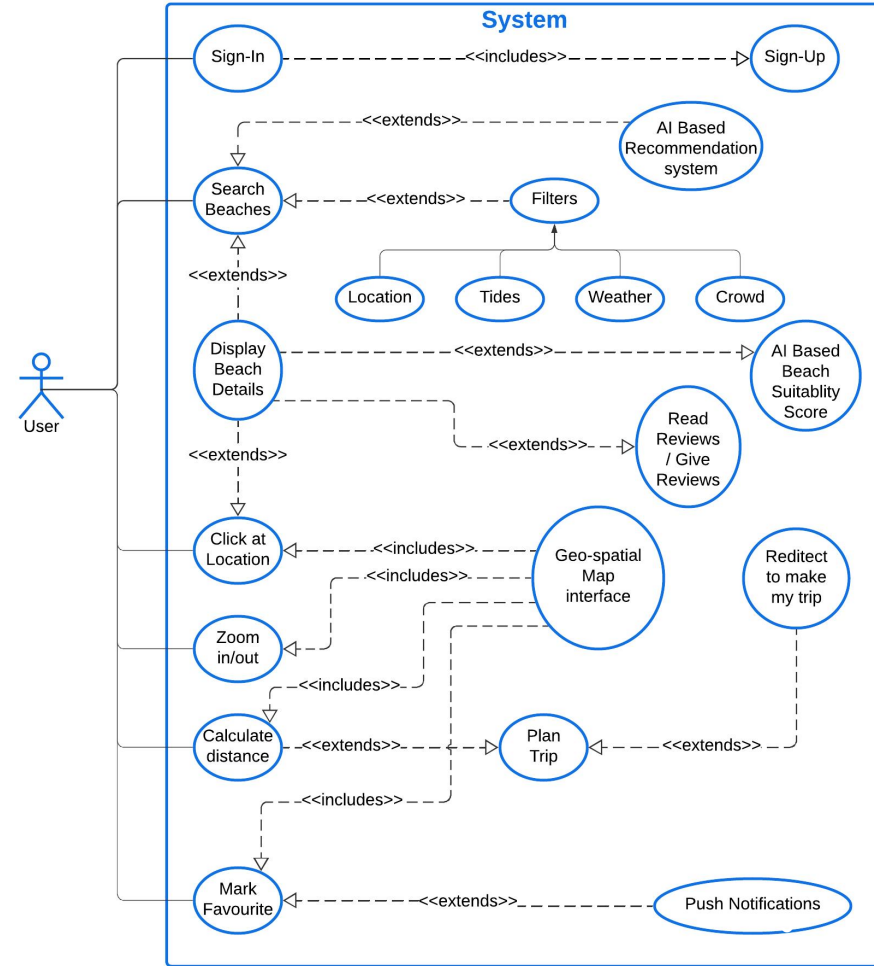
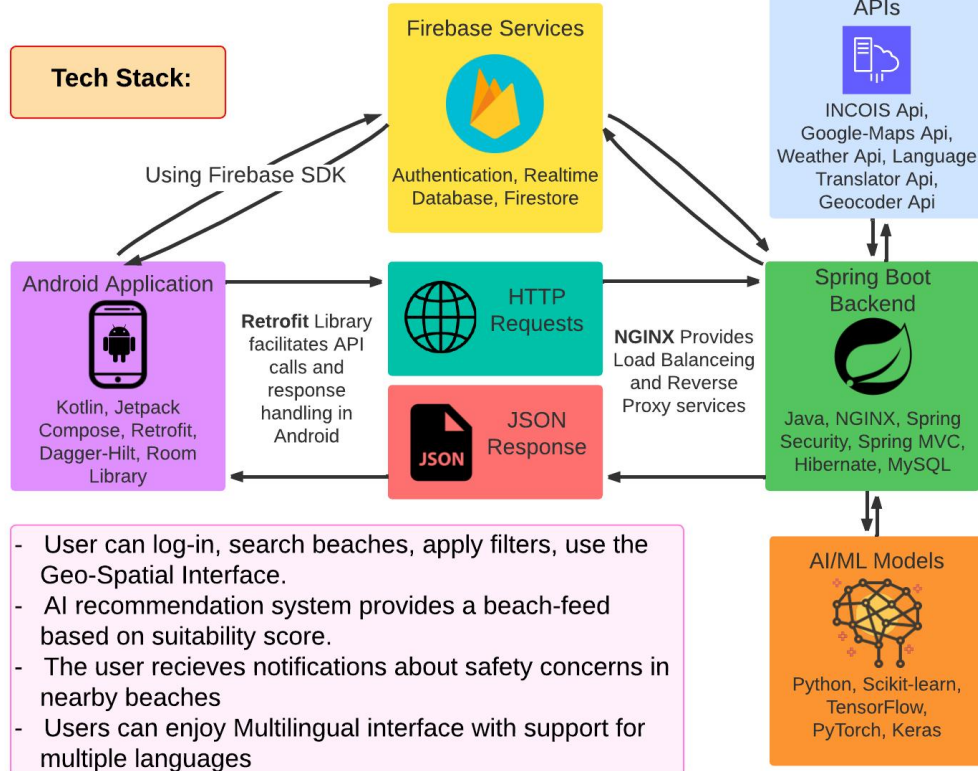




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TECHNICAL APPROACH

★ TechStack (diagram 1) & Use-Cases (diagram 2):





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FEASIBILITY AND VIABILITY



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★ Analysis of the feasibility of the idea

- Technology: The application uses reliable, scalable and widely accepted technologies.
- All the data sources, such as APIs, are provided and maintained by the Government of India.
- The app can generate revenue through ads and partnerships with coastal tourism businesses.

★ Potential challenges and risks

- The server may go down due to excessive number of requests
- Providing an accurate data for dynamically changing environment
- Security principles for the application and user data.
- Showing geospatial maps in an interactive way.

★ Strategies for overcoming these challenges

- Use of NGINX for reverse proxy, load balancing and caching.
- Use of ML - based models and standard API's for fetching the data.
- Encrypted user credentials using firebase and Rest API security using JWT tokens.
- Use of ArcGIS for geospatial map visualization.



ArcGIS



NGINX



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IMPACT AND BENEFITS



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★ Potential impact on the target audience

- **Enhanced Tourist Safety:** Reduces the likelihood of accidents and fatalities by providing real-time updates on beach conditions.
- **Support for Blue Economy:** Contributes to the growth of coastal tourism in line with India's Blue Economy policy.
- **Informed Decision-Making:** Empowers tourists with accurate data for better planning of recreational activities at beaches.

★ Benefits of the solution

- **Real-Time Alerts:** Provides instant notifications of hazardous conditions like high waves or strong winds, improving user safety.
- **Geospatial Visualization:** Offers easy-to-understand maps with color-coded suitability ratings for beach locations.
- **AI-Generated Suitability Scores:** Assigns AI-driven suitability scores to each beach based on multiple factors, helping users quickly identify the best locations for recreational activities.



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RESEARCH AND REFERENCES



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Reference:

1. [Indian National Centre for Ocean Information Services \(INCOIS\)](#)
2. [Draft Blue Economy Policy of India](#)

Citations:

1. Goodchild, M. F., & Li, L. (2012). Assuring the quality of volunteered geographic information. Spatial Statistics, 1, 110-120. <https://doi.org/10.1016/j.spasta.2012.03.001>
2. Li, X., Zhang, W., & Yang, J. (2021). Application of artificial intelligence in environmental monitoring and management: A review. Journal of Environmental Management, 293, 112877. <https://doi.org/10.1016/j.jenvman.2021.112877>

